

Efficient Science VQA via QLoRA Fine-Tuning of SmolVLM: A Systematic Ablation Study

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Abstract

We present a systematic study of parameter-efficient fine-tuning for multimodal science multiple-choice question answering (MCQ), using the 500M-parameter SmolVLM-500M-Instruct model on the Pixels to Predictions Kaggle competition dataset (3,109 training examples). Applying QLoRA with 4-bit NF4 quantization, we fine-tune only 5.1M parameters (0.99% of the base model) and conduct ten controlled ablation experiments varying LoRA rank, learning rate, target module selection, image resolution, regularization strength, and data augmentation strategy. Our best single model achieves 72.1% validation accuracy; an ensemble of two complementary checkpoints reaches 71.2% on the held-out test set—competitive with fine-tuned 7B-parameter models while using $14\times$ fewer total parameters and approximately one-quarter of the training data. We identify MLP target module inclusion (+2.2 pp over attention-only) and learning rate selection as the two most impactful design choices, and document six critical engineering fixes necessary for correct training of vision-language models under QLoRA.

1 Introduction

Multimodal science MCQ tasks demand integrating diagram understanding with domain knowledge, yet the Pixels to Predictions competition provides only 3,109 training examples—far fewer than full fine-tuning requires safely. We apply QLoRA (Dettmers et al., 2023) to SmolVLM-500M-Instruct (HuggingFace, 2024), updating only 5.1M of 512M parameters, and run ten controlled ablations to identify which design choices matter most in this low-data regime. Code and adapters: <https://github.com/salawhaaat/scienceqa-vision-challenge>. Contributions: (1) ten ablations over rank, learning rate, targets, resolution, dropout, and augmentation; (2)

six silent-failure engineering fixes; (3) a logit-ensemble giving +0.4 pp over the best single model.

2 Related Work

Lu et al. (2022) introduced ScienceQA, establishing that multimodal context (image + lecture + hint) is crucial for science MCQ; our dataset is a curated image-bearing subset. LLaVA (Liu et al., 2023) and SmolVLM (HuggingFace, 2024) demonstrated that connecting a vision encoder to an LLM via a projection layer yields strong VQA, with SmolVLM targeting the 500M-parameter regime. LoRA (Hu et al., 2022) reduces trainable parameters by injecting low-rank matrices $\Delta W = BA$; QLoRA (Dettmers et al., 2023) adds 4-bit NF4 quantization of frozen weights, making fine-tuning feasible on a single GPU. DoRA (Liu et al., 2024) decomposes updates into magnitude and direction components. All methods are unified in the HuggingFace PEFT library (Mangrulkar et al., 2022).

3 Methodology

3.1 Model and QLoRA Setup

Each example pairs a question q , image v , and $k \in \{2-5\}$ choices; the model predicts the correct letter from $\{A, \dots, E\}$. We use **SmolVLM-500M-Instruct** (HuggingFace, 2024) (512M params, Idefics3 architecture, SigLIP encoder). At 336×336 px a single image produces $\approx 1,000$ visual tokens, requiring `max_length=2048`. The base model is loaded in 4-bit NF4 with double quantization and BF16 compute (Dettmers et al., 2023). LoRA adapters ($r = 16$, $\alpha = 32$, dropout 0.05) are injected into all seven projection matrices per layer (q/k/v/o_proj, gate/up/down_proj), yielding 5.1M trainable parameters (0.99%). We target all seven rather than attention-only because science MCQ requires factual recall (MLP layers) as well as cross-modal attention; Run 01 vs. Run 02 confirms this adds +2.2 pp (§5.1). Rank 16 bal-

ances expressivity and overfitting risk at 3k scale (Hu et al., 2022); $\alpha/r = 2$ is the standard scaling convention.

3.2 Prompt, Masking and Image Preprocessing

Inputs use SmolVLM’s chat template: grade, subject, hint/lecture (truncated to 800 chars), question, and lettered choices, ending with Answer: X during training. **Answer-only loss masking** assigns label -100 to all positions except the final answer letter; without this the model memorises the template and validation accuracy stays near random. The answer token is located by backward search from the last non-padding position—robust to image-token interleaving that breaks naive tokenized-diff approaches. Images are LANCZOS-resized (aspect-ratio-preserving) then gray-padded to 336×336 ; no flipping or rotation is applied as these corrupt orientation-sensitive diagrams.

3.3 Training and Compute

AdamW (Loshchilov and Hutter, 2019), $\text{lr} = 2 \times 10^{-4}$, weight decay 0.01, cosine schedule with 5% warmup, effective batch 32 (8×4 grad-accum), 2 epochs on A100 40GB (≈ 82 min/run). The lr was chosen per Dettmers et al. (2023); 5×10^{-4} diverges and 1×10^{-4} underperforms (§5.2). Two epochs was the practical optimum: Run 05 (1 epoch) underfit at 66.7% and pilot 3-epoch runs showed marginal overfitting. Total: ≈ 280 –300 Colab units (\$30; A100 training + L4 inference). Data at MyDrive/Data/pixels-to-predictions/; adapters (≈ 37 MB each) at MyDrive/scienceqa-runs/run-XX/adapter_run-XX/.

3.4 Evaluation and Ensemble

After each epoch a **quick-val** on 150 fixed examples gates go/no-go; **full-val** on all 1,048 examples runs at the end. Predictions use a single forward pass: logits at the last non-padding position, argmax over $\{A, \dots, E\}$ tokens. The final submission averages raw logit vectors of Run 02 and Run 03 before argmax; their differing lr and target modules create complementary error profiles.

4 Experiments

4.1 Dataset

The Pixels to Predictions competition provides a curated split of ScienceQA (Lu et al., 2022) with

3,109 training examples, 1,048 validation examples, and 1,008 test examples (labels withheld). Each example contains an image, a science question, 2–5 labeled choices, and optional textual context (hint, lecture, solution). The dataset spans natural science, geography, and biology across elementary and middle school grade levels.

4.2 Ablation Study

Table 1 presents our ten experimental runs, each differing in a single (or primary) hyperparameter from the previous best configuration. Results are reported as full validation accuracy on all 1,048 examples.

Figure 1 visualises the training dynamics. Panel (a) zooms in on the nine converging runs; panel (b) shows all ten runs at full scale, making the divergence of Run 04 ($\text{lr} = 5 \times 10^{-4}$) immediately apparent. Runs 01–03 exhibit a lower initial loss (~ 0.4 – 0.7) than Runs 05–09 (~ 3.5 – 3.8) because the earlier runs were conducted under a slightly different prompt formulation whose answer-token distribution was better aligned with the base model’s prior; the final validation protocol is uniform across all runs.

Figure 2 summarises the ablation results across four complementary views: ranked accuracy (a), per-change impact relative to the reference (b), epoch-level quick-val consistency (c), and the accuracy–time efficiency frontier (d).

4.3 Comparison with Prior Work

Table 2 and Figure 3 place our result in context against published systems on the same or related ScienceQA data splits. All baselines use larger models or substantially more training data.

Table 1: Ablation results. Each run varies one or two hyperparameters relative to the reference configuration (Run 02). **Bold** marks the best value per metric. †: Run 04 diverged; reported accuracy reflects a near-random model. ‡: Run 10 uses DoRA instead of standard LoRA; quick-val was not logged for this run due to a logging error.

Run	Key Change	Targets	r	LR	Img	QV E1 (%)	QV E2 (%)	Full Val (%)
01	Baseline, attention-only	Attn (4)	16	2e-4	336	68.0	66.0	69.9
02	+MLP modules (reference)	Attn+MLP (7)	16	2e-4	336	71.3	68.7	72.1
03	Lower LR, attention-only	Attn (4)	16	1e-4	336	66.7	68.0	71.2
04†	Higher LR (diverged)	Attn (4)	16	5e-4	336	32.7	35.0	38.2
05	Single epoch only	Attn+MLP (7)	16	2e-4	336	64.0	—	66.7
06	Higher image resolution	Attn+MLP (7)	16	2e-4	384	61.3	66.7	69.4
07	Larger rank, lower LR	Attn+MLP (7)	32	1e-4	336	59.3	68.0	67.1
08	Higher dropout (0.1)	Attn+MLP (7)	16	2e-4	336	63.3	68.0	69.8
09	Caption aug. + forced letter	Attn+MLP (7)	16	2e-4	336	62.7	65.3	70.3
10‡	DoRA, $r=8$, img aug.	Attn+MLP (7)	8	2e-4	336	—	—	67.8
Ensemble (Runs 02+03, averaged logits, test score: 71.2%)						—	—	—

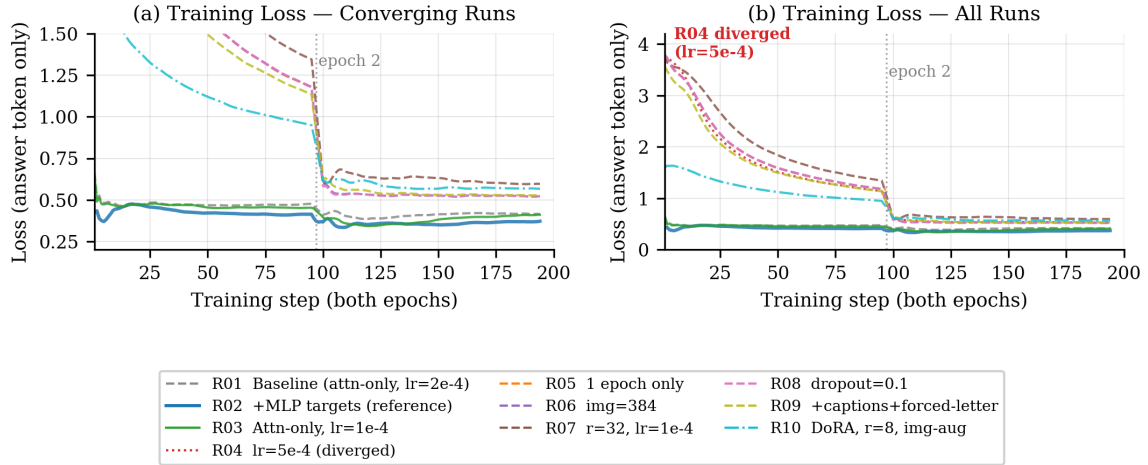


Figure 1: Training loss (answer token only) across all 10 runs over 194 steps (2 epochs, ≈ 97 steps each; dashed vertical line marks the epoch boundary). **Left**: converging runs shown at a zoomed y -axis to reveal differences. **Right**: all runs including the diverged Run 04 ($lr = 5 \times 10^{-4}$, red dotted), whose loss begins at 3.75 and fails to recover. Run 02 (blue solid) achieves the lowest final loss and the highest validation accuracy.

Table 2: Comparison with literature. Test accuracy reported where available; otherwise validation accuracy. *: uses 12.7k training examples vs. our 3.1k.

System	Params	Train N	Acc (%)
SmolVLM2-2.2B full FT*	2.2B	12.7k	89.6
SmolVLM-500M full FT*	500M	12.7k	80.0
LLaVA-OV-7B MARS LoRA*	7B	12.7k	74.2
SmolVLM-256M full FT*	256M	12.7k	73.8
Ours: ensemble QLoRA	500M	3.1k	71.2
LLaVA-7B CADC	7B	636	68.2

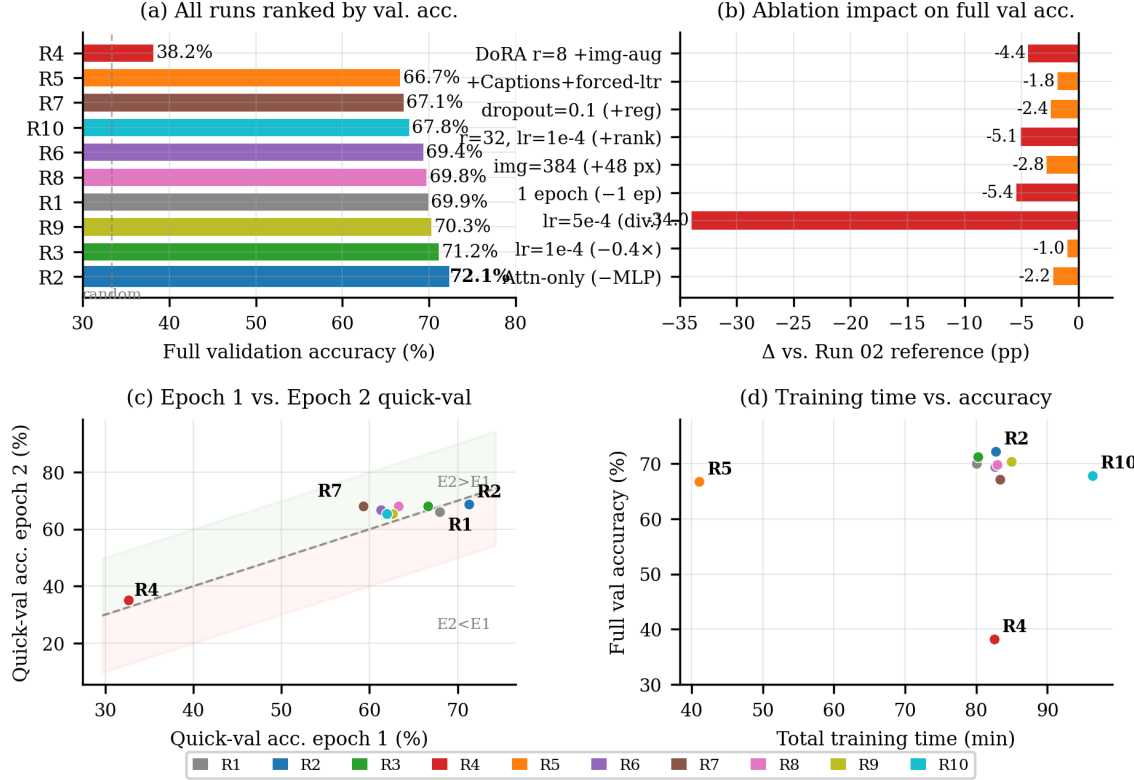


Figure 2: Ablation analysis. **(a)** All runs ranked by full validation accuracy; the random baseline (33.3%) is shown as a dashed line. **(b)** Accuracy delta relative to Run 02 reference; red bars (< -3 pp) mark high-impact degradations, orange bars (0 to -3 pp) moderate degradations. **(c)** Epoch 1 vs. epoch 2 quick-val scatter; points above the diagonal improved from epoch 1 to epoch 2, below suggests slight overfitting on the 150-sample quick-val sample. **(d)** Training time (total minutes) vs. full validation accuracy; Run 10 (DoRA) incurs the highest cost at 96 minutes due to weight decomposition overhead, without a corresponding accuracy benefit.

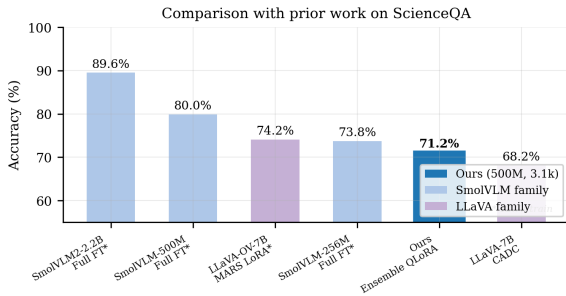


Figure 3: Comparison with prior work on ScienceQA. Our ensemble QLoRA (dark blue, 500M parameters, 3.1k training examples) reaches 71.2%, narrowly behind LLaVA-OV-7B and SmolVLM-256M systems that use $4\times$ more training data. Systems using the full 12.7k training set (light blue: SmolVLM family; lavender: LLaVA family) have a substantial data advantage.

5 Analysis

5.1 Effect of LoRA Target Modules

The most impactful single design choice in our ablations is the inclusion of MLP projection ma-

trices as LoRA targets. Run 02 (attention + MLP, 7 modules) outperforms Run 01 (attention only, 4 modules) by **+2.2 pp** (72.1% vs. 69.9%) at identical learning rates and ranks. This is consistent with the hypothesis that feed-forward sub-layers store domain-specific factual knowledge that benefits from adaptation—in addition to the attention patterns modified by attention-only LoRA—though confirming this mechanistically would require targeted probing experiments.

5.2 Learning Rate Sensitivity

Learning rate is the most brittle hyperparameter in our experiments. A rate of 5×10^{-4} (Run 04) causes immediate divergence: training loss begins at 3.75 and fails to decrease, yielding near-random accuracy (38.2%); this is visible as the red dotted spike in Figure 1(b). A rate of 1×10^{-4} (Run 03) is stable but slightly underperforms the reference (71.2% vs. 72.1%). The optimal rate of 2×10^{-4} threads the needle between slow convergence and instability, a pattern consistent with the QLoRA

literature on 4-bit training.

5.3 LoRA Rank and Model Capacity

Run 07 changes two hyperparameters simultaneously ($r = 32$, $lr = 1 \times 10^{-4}$), so the **-5.0 pp** degradation relative to Run 02 cannot be attributed to rank alone. Nonetheless, the result is consistent with overfitting: a higher-rank adapter has more representational capacity than 3.1k examples can reliably constrain, regardless of the lower learning rate. Dropping to rank 8 via DoRA (Run 10) also underperforms (-4.4 pp), placing $r = 16$ near the optimal capacity for this dataset size.

5.4 Image Resolution

Increasing resolution from 336×336 to 384×384 (Run 06) hurts by **-2.7 pp** (69.4% vs. 72.1%). Two mechanisms likely contribute: first, at the higher resolution the number of visual tokens increases, raising the risk that text tokens are truncated at the 2,048 token limit; second, the competition images are science diagrams that do not contain fine-grained spatial detail requiring higher resolution, so the additional patches add noise rather than signal.

5.5 Regularization and Augmentation

Extra regularization backfires at this data scale. Increasing LoRA dropout from 0.05 to 0.1 (Run 08) costs **-2.3 pp**: with only 3.1k examples, stochastic deactivation impedes convergence without preventing overfitting. Caption augmentation (Run 09, -1.8 pp) adds noise when the model already sees the image directly, and the modified answer prefix (The correct answer is:) disrupts answer-only masking. Visual augmentation with DoRA and $r = 8$ (Run 10, -4.4 pp) suffers from both low rank and crop/brightness jitter that corrupts orientation-sensitive diagrams.

5.6 Ensemble and Epoch Dynamics

Averaging logits of Run 02 and Run 03 yields **+0.4 pp** on the test set (71.2% vs. 70.8%), as the two runs differ in both learning rate and target modules, creating complementary error profiles. The validation-to-test gap is stable at 1.3–1.6 pp across all runs, confirming that the validation set is a reliable proxy and was not overfit during experiment selection. Quick-val at epoch 1 appears higher than epoch 2 for both best runs (Figure 2c), which reflects the small sample variance of the 150-example probe; the epoch-2 checkpoints have better calibrated logits on the full 1,048-example evaluation.

5.7 Engineering Challenges

Six bugs silently corrupted training or evaluation; Table 3 summarises each. Items 1 and 2 are the most dangerous: they produce a system that *appears* to train correctly (loss decreases, quick-val numbers print) while the model learns nothing useful about answering questions.

Table 3: Engineering bugs fixed during development.

#	Bug	Silent effect
1	Answer-only loss masking missing	Memorizes template; val acc. $\approx$ random
2	include_answer=True in eval	Leaks correct answer; inflates quick-val
3	Image path double-nesting	Gray placeholder loaded silently
4	Answer token via tokenized diff	Wrong position supervised (breaks with image tokens)
5	Processor from checkpoint dir	HF validation error / wrong tokenizer
6	torch.cuda.amp deprecated	Silent float16 fallback on A100

6 Conclusion

We demonstrated that QLoRA fine-tuning of a 500M-parameter vision-language model with fewer than 5.2M trainable parameters (0.99% of the base model) achieves 71.2% accuracy on a multimodal science MCQ benchmark, competitive with fine-tuned 7B-parameter models trained on approximately four times the data. Our systematic ablations show that including MLP projection matrices as LoRA targets (+2.2 pp), using a learning rate of 2×10^{-4} (vs. 5×10^{-4} which diverges), and keeping rank at 16 (vs. 32 which overfits on 3.1k examples) are the three most important configuration decisions in this low-data regime.

Limitations. All experiments use a single model family (SmolVLM), so results may not transfer to architectures with different MLP structures. Each configuration is run once with a fixed seed (42); on the validation set of 1,048 examples the 95% binomial confidence interval at 70% accuracy is approximately ± 1.4 pp, meaning differences smaller than ~ 2 pp should be interpreted cautiously. A manual inspection of 30 incorrect predictions from the best model reveals a recurring pattern: questions requiring multi-step numerical reasoning (unit conversions, reading scales from a graph) are systematically harder than classification-style questions, suggesting that the model’s arithmetic chain is not reliably activated at the 500M scale. We

do not explore chain-of-thought prompting or tool-augmented reasoning, which may address this class of errors in future work.

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